IGCSE 0607 Practice Questions — E10.1 — Combined Questions & Markscheme

Classifying statistical data

Total Questions: 18 — **Total Marks:** 127 Marks

Table of Contents & Quick Navigation

- **E10.1** Classifying statistical data
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(10 Qs)*

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1 70 students are asked to choose their favourite sport from Football, Tennis, or Basketball.
The incomplete two-way table shows some information about their choices.

(a) Complete the two-way table.

	Football	**Tennis**	**Basketball**	**Total**
Boys	15		12	
Girls		14		30
Total	23		20	70

[3]

(b) A student is chosen at random.
Find the probability that the student is a boy whose favourite sport is Tennis.

..... **[1]**

2 50 students each choose one sport to play: Tennis, Football, or Swimming.
The students are classified as Boys or Girls.

(a) Complete the two-way table to show this information.

	Tennis	**Football**	**Swimming**	**Total**
Boys	7			28
Girls	8		6	
Total		20		50

[3]

(b) A student is chosen at random.
Find the probability that the student is a boy who chose Football.

..... **[1]**

3 A group of 50 students were asked which sport they prefer.
Some information is shown in the two-way table.

(a) Complete the two-way table.

	Football	Tennis	Swimming	Total
Boys	15	8		28
Girls		8		
Total	20			50

[3]

(b) A student is chosen at random.
Find the probability that the student is a girl who prefers Swimming.

..... **[1]**

4 50 students were asked to choose their favourite sport from Football, Basketball, and Tennis.
The two-way table shows some information about their choices.

(a) Complete the two-way table.

	Football	**Basketball**	**Tennis**	**Total**
Boys	15		5	28
Girls		10	6	
Total	21			50

[3]

(b) A student who chose Tennis is chosen at random.
Work out the probability that the student is a boy.

..... **[2]**

5 80 students were asked to choose their favourite sport from Football, Basketball, and Tennis.

- There are 35 girls in total.
- 38 students chose Football, of which 24 were boys.
- 12 girls chose Basketball.
- 20 students chose Tennis.

(a) Complete the two-way table.

	Football	**Basketball**	**Tennis**	**Total**
Boys				
Girls				
Total				

[3]

(b) A student is chosen at random.
Find the probability that the student is a boy whose favourite sport is Basketball.

..... **[1]**

(c) A girl is chosen at random.
Find the probability that her favourite sport is Tennis.

..... **[2]**

6 A fitness instructor collects statistical data from 24 members of a sports club.

(a) Classify each of the following types of data as qualitative, discrete, or continuous:

(i) favourite sport,

..... **[1]**

(ii) height, in metres,

..... **[1]**

(iii) number of hours spent exercising each week, rounded to the nearest integer.

..... **[1]**

(b) The resting heart rates, r beats per minute, of the 24 members are listed below.

64	72	58	81	67	75	60	78	54	69	84	71
63	77	59	68	70	82	65	74	56	66	73	80

Complete the grouped tally and frequency table below.

Resting heart rate (r)	Tally	Frequency
$50 \leq r < 60$	||||	4
$60 \leq r < 70$		
$70 \leq r < 80$		
$80 \leq r < 90$		

[3]

(c) Calculate the percentage of these 24 members who have a resting heart rate of less than 70 beats per minute.

..... % **[2]**

7 A survey of adults recorded their preferred hot drink, either Tea or Coffee.
Let x be the number of males who prefer Coffee.
The number of males who prefer Tea is $2x + 10$.
The number of females who prefer Coffee is $x + 5$.
The number of females who prefer Tea is $3x - 11$.

(a) Write down an expression, in terms of x, for the total number of adults.
Give your answer in its simplest form.

..... **[2]**

(b) The survey included 130 adults.
Find the value of x.

$x =$ ..... **[2]**

(c) Complete the two-way table.

	Coffee	**Tea**	**Total**
Male			
Female			
Total			

[3]

(d) One of the adults who prefers Coffee is chosen at random.
Find the probability that this adult is female.

..... **[2]**

8 A researcher classifies 150 participants in a clinical trial according to their blood group (A, B, or O) and whether they responded positively (Yes) or did not respond positively (No) to a new treatment. The two-way table below shows the number of participants in each category, where x and y are positive integers.
The total number of participants with blood group B is 28.

	Blood group A	Blood group B	Blood group O
Yes	$x^2 + 2x$	$2x + y$	$4x - y$
No	$2x + 3y$	$x + 6$	$3x + 3y$

(a) **(i)** Use the total number of participants with blood group B to express y in terms of x.

$y =$ ..... **[2]**

(ii) Show that $x^2 - 4x - 12 = 0$.

[3]

(b) Solve the equation $x^2 - 4x - 12 = 0$ to find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[3]**

(c) **(i)** Complete the numerical two-way table below using your values of x and y.

	Group A	Group B	Group O	Total
Yes				
No				
Total		28		150

[2]

(ii) Calculate the percentage of all 150 participants who responded positively (Yes) to the treatment.

..... % **[2]**

9 A group of 120 adults were asked whether they prefer Tea or Coffee.

- 55 of the adults are women.
- 70 of the adults prefer Tea.
- 25 of the men prefer Coffee.

(a) Complete the two-way table to show this information.

	Tea	**Coffee**	**Total**
Men			
Women			
Total			

[3]

(b) Work out the number of women who prefer Tea as a percentage of the total number of adults.

..... % **[2]**

10 A company has 120 employees. There are 75 men and 45 women.
They travel to work by bus, train, or car.

- 20% of the men travel by bus.
- $\frac{1}{3}$ of the women travel by bus.
- A total of 38 employees travel by train, of which 14 are women.
- The remaining employees travel by car.

(a) Construct and complete the two-way table to show this information.

	Bus	**Train**	**Car**	**Total**
Men				
Women				
Total				

[4]

(b) An employee who travels by car is chosen at random.
Work out the probability that this employee is a man.

..... **[2]**

11 In a school of 400 students, the ratio of juniors to seniors is 3 : 2.
Each student travels to school by Bus, by Car, or Walks.

- 45% of the 400 students take the Bus.
- Of the seniors, 30 take the Car and 50 Walk.
- There are 80 juniors who take the Car.

(a) Complete the two-way table to classify this data.

	Bus	**Car**	**Walk**	**Total**
Juniors				
Seniors				
Total				400

[3]

(b) A student is chosen at random.
Find the probability that they are a junior who walks.

..... **[1]**

(c) Two seniors are chosen at random.
Find the probability that they both take the bus.

..... **[2]**

12 A group of 85 people were asked about their favourite drink.
The two-way table shows the number of people in terms of x.

	Tea	Coffee	Water
Men	x	$2x - 5$	$x + 6$
Women	$x - 2$	$x + 5$	$2x + 1$

(a) Find the value of x.

$x =$ ..... **[3]**

(b) Find the probability that a person chosen at random is a man who prefers Water.

..... **[2]**

(c) A person who prefers Coffee is chosen at random.
Find the probability that this person is a woman.

..... **[2]**

13 A school surveys 300 students in Year 9, Year 10, and Year 11 about their main method of travel to school: Walk, Bus, or Car.

- 32% of the 300 students are in Year 9.
- There are 18 more students in Year 10 than in Year 9.
- $\frac{3}{8}$ of the Year 9 students travel by Bus, and 25 Year 9 students travel by Car.
- Among the Year 11 students, the ratio of those who Walk to those who travel by Bus is 5 : 7.

(a) Complete the two-way table below to show the results of the survey.

	Walk	Bus	Car	Total
Year 9				
Year 10				
Year 11			18	
Total	115		85	300

[5]

(b) Calculate the percentage of Year 10 students who travel to school by Car, giving your answer correct to 1 decimal place.

..... % **[2]**

(c) One of the students who travels by Bus is chosen at random.
Find the probability, as a fraction in its simplest form, that this student is in Year 11.

..... **[2]**

14 80 students travel to school either by Bus or by Walking.
The two-way table shows the number of boys and girls using each method of transport, in terms of x.

	Bus	**Walk**	**Total**
Boys	x	$2x$	
Girls	$x + 5$	$3x - 2$	
Total			80

(a) Work out the value of x.

$x =$ ..... **[3]**

(b) Complete the two-way table with the number of students in each category.

	Bus	**Walk**	**Total**
Boys			
Girls			
Total			80

[2]

(c) A student is chosen at random from those who walk to school.
Find the probability that this student is a girl.

..... **[2]**

15 300 students each choose exactly one activity: Art, Music, or Drama.

- The ratio of boys to girls is 2 : 3.
- 20% of the boys choose Art.
- The number of boys who choose Drama is twice the number of boys who choose Art.
- $\frac{1}{3}$ of the girls choose Drama.
- The ratio of girls who choose Art to girls who choose Music is 3 : 5.

(a) Complete the two-way table to show this information.

	Art	Music	Drama	Total
Boys				
Girls				
Total				

[5]

(b) A student is chosen at random.
Find the probability that the student is a girl who does not choose Drama.

..... **[2]**

16 A company has 300 employees. Each employee is either a Manager, an Office Worker, or a Technician.
Each employee works either Full-Time or Part-Time.

- The ratio of Managers : Office Workers : Technicians is 2 : 5 : 3.
- 20% of the Managers work Part-Time.
- 40% of the Office Workers work Part-Time.
- The number of Part-Time Technicians is equal to the number of Part-Time Managers.

(a) Complete the two-way table for this information.

	Full-Time	**Part-Time**	**Total**
Managers			
Office Workers			
Technicians			
Total			

[4]

(b) A Part-Time employee is chosen at random.
Find the probability that they are an Office Worker.

..... **[2]**

(c) Two Managers are chosen at random.
Find the probability that exactly one of them works Part-Time.

..... **[3]**

17 A sports club has 200 members. Each member plays either Golf or Tennis.
There are x male members in the club.

- The number of male members who play Golf is $\frac{1}{4}x$.
- The number of female members who play Tennis is 30 more than the number of male members who play Golf.
- A total of 110 members play Tennis.

(a) Write down an expression, in terms of x, for the number of male members who play Tennis.
Give your answer in its simplest form.

..... **[1]**

(b) Use the information about the members who play Tennis to form an equation in x, and show that $x = 80$.

[2]

(c) Complete the two-way table to show the number of members in each category.

	Golf	**Tennis**	**Total**
Male			
Female			
Total			

[3]

(d) Two members are chosen at random from those who play Golf.
Calculate the probability that they are both female.

..... **[3]**

18 A group of 120 students are classified by gender (Boys, Girls) and handedness (Left-handed, Right-handed).
The two-way table shows the number of students in each category in terms of x and y.
Two students are chosen at random from the 120 students.
The probability that both students are left-handed is $\frac{3}{34}$.

	Left-handed	**Right-handed**
Boys	x	y
Girls	$2x$	$y - 6$

(a) Show that $3x^2 - x - 420 = 0$.

[4]

(b) Find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(c) A right-handed student is chosen at random.
Find the probability that the student is a girl.

..... **[2]**

1 **(a)** Boys Total = 40 and Girls Football = 8 **[B1]**

Girls Basketball = 8 and Total Tennis = 27 **[B1]**

Boys Tennis = 13 **[B1]**

(b) $\frac{13}{70}$ **[B1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** Tennis Total = 15, Girls Total = 22 **[B1]**

Boys Football = 12, Boys Swimming = 9 **[B1]**

Girls Football = 8 **[B1]**

(b) $\frac{12}{50}$ or $\frac{6}{25}$ **[B1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** 5 in Boys/Swimming cell **[B1]**

5 in Girls/Football, 9 in Girls/Swimming, 22 in Girls/Total **[B1]**

16 in Total/Tennis, 14 in Total/Swimming **[B1]**

(b) $\frac{9}{50}$ **[B1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Boys Basketball = 8 **[B1]**

Girls Football = 6 AND Girls Total = 22 **[B1]**

Total Basketball = 18 AND Total Tennis = 11 **[B1]**

(b) Denominator of 11 seen or fraction $\frac{5}{n}$ where $n > 5$ **[M1]**

$\frac{5}{11}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** 6 or more correct missing entries **[B3]**

4 or 5 correct missing entries **[B2]**

2 or 3 correct missing entries **[B1]**

(Correct Boys row: 24, 10, 11, 45)

(Correct Girls row: 14, 12, 9, 35)

(Correct Total row: 38, 22, 20, 80)

(b) $\frac{10}{80}$ or $\frac{1}{8}$ **[B1]**

(c) Denominator of 35 seen **[M1]**

$\frac{9}{35}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

6 (a) (i) Qualitative **[B1]**

(ii) Continuous **[B1]**

(iii) Discrete **[B1]**

(b) Row $60 \leq r < 70$: correct tally of 8 and frequency 8 **[B1]**

Row $70 \leq r < 80$: correct tally of 8 and frequency 8 **[B1]**

Row $80 \leq r < 90$: correct tally of 4 and frequency 4 **[B1]**

(c) $\frac{4+8}{24} \times 100$ **[M1]**

50 **[A1]**

[Back to Question] *[Total Marks: 8]*

7 (a) $x + (2x + 10) + (x + 5) + (3x - 11)$ **[M1]**

$7x + 4$ **[A1]**

(b) $7x + 4 = 130$ **[M1]**

$x = 18$ **[A1]**

(c) Correct Male row: 18, 46, 64 **[B1]**

Correct Female row: 23, 43, 66 **[B1]**

Correct Total row: 41, 89, 130 **[B1]**

(d) Denominator of 41 seen **[M1]**

$\frac{23}{41}$ **[A1]**

[Back to Question] *[Total Marks: 9]*

8 (a) (i) $(2x + y) + (x + 6) = 28$ **[M1]**

$22 - 3x$ **[A1]**

(ii) $(x^2 + 2x) + (2x + y) + (4x - y) + (2x + 3y) + (x + 6) + (3x + 3y) = 150$ simplifying to $x^2 + 14x + 6y + 6 = 150$ **[M1]**

$x^2 + 14x + 6(22 - 3x) - 144 = 0$ **[M1]**

$x^2 + 14x + 132 - 18x - 144 = 0$ simplifying to $x^2 - 4x - 12 = 0$ **[A1]**

(b) $(x - 6)(x + 2) = 0$ **[M1]**

$x = 6$ (rejecting $x = -2$ since $x > 0$) **[A1]**

$y = 22 - 3(6) = 4$ **[A1]**

(c) **(i)** Row *Yes*: $48, 16, 20$, Total $= 84$ **[B1]**

Row *No*: $24, 12, 30$, Total $= 66$; Column Totals: $72, 28, 50$ **[B1]**

(ii) $\frac{84}{150} \times 100$ **[M1]**

56 **[A1]**

[Back to Question] *[Total Marks: 12]*

9 **(a)** Total Women $= 55$, Total Tea $= 70$, Overall Total $= 120$ placed correctly **[B1]**

Men Coffee $= 25$ and Total Coffee $= 50$ or Men Total $= 65$ placed correctly **[B1]**

Men Tea $= 40$, Women Tea $= 30$, Women Coffee $= 25$ **[B1]**

(b) $\frac{30}{120} \times 100$ **[M1]**

25 **[A1]**

[Back to Question] *[Total Marks: 5]*

10 **(a)** Total column correct: 75 (Men), 45 (Women), 120 (Total) **[B1]**

Bus column correct: 15 (Men), 15 (Women), 30 (Total) **[B1]**

Train column correct: 24 (Men), 14 (Women), 38 (Total) **[B1]**

Car column correct: 36 (Men), 16 (Women), 52 (Total) **[B1]**

(b) Denominator of 52 seen or fraction $\frac{36}{n}$ **[M1]**

$\frac{36}{52}$ or $\frac{9}{13}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** Total Juniors $= 240$, Total Seniors $= 160$ **[B1]**

Total Bus $= 180$, Total Car $= 110$, Total Walk $= 110$ **[B1]**

Juniors: Bus $= 100$, Car $= 80$, Walk $= 60$ AND Seniors: Bus $= 80$, Car $= 30$, Walk $= 50$ **[B1]**

(b) $\frac{60}{400}$ or $\frac{3}{20}$ **[B1]**

(c) $\frac{80}{160} \times \frac{79}{159}$ **[M1]**

$\frac{79}{318}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

12 **(a)** $x + 2x - 5 + x + 6 + x - 2 + x + 5 + 2x + 1 = 85$ or $8x + 5 = 85$ **[M1]**

$8x = 80$ **[M1]**

10 **[A1]**

(b) Man who prefers Water $= 10 + 6 = 16$ **[M1]**

$\frac{16}{85}$ **[A1]**

(c) Total who prefer Coffee $= 15 + 15 = 30$ **[M1]**

$\frac{15}{30}$ or $\frac{1}{2}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

13 **(a)** Row totals: Year 9 = 96, Year 10 = 114, Year 11 = 90 **[B1]**

Column total Bus = 100 **[B1]**

Year 9 row completed correctly: Walk = 35, Bus = 36, Car = 25 **[B1]**

Year 11 row completed correctly: Walk = 30, Bus = 42 **[B1]**

Year 10 row completed correctly: Walk = 50, Bus = 22, Car = 42 **[B1]**

(b) $\frac{42}{114} \times 100$ $(= 36.842\ldots)$ **[M1]**

36.8 **[A1]**

(c) $\frac{42}{100}$ **[M1]**

$\frac{21}{50}$ **[A1]**

[Back to Question] *[Total Marks: 9]*

14 **(a)** $x + 2x + (x + 5) + (3x - 2) = 80$ **[M1]**

$7x + 3 = 80$ **[M1]**

$x = 11$ **[A1]**

(b) Boys Bus = 11, Boys Walk = 22, Boys Total = 33 **[B1]**

Girls Bus = 16, Girls Walk = 31, Girls Total = 47 and Column totals (Bus = 27, Walk = 53) correct **[B1]**

(c) Denominator of 53 seen **[M1]**

$\frac{31}{53}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

15 **(a)** Boys total = 120 and Girls total = 180 **[B1]**

Boys/Art = 24 **[B1]**

Boys/Drama = 48 and Boys/Music = 48 **[B1]**

Girls/Drama = 60 **[B1]**

Girls/Art = 45 and Girls/Music = 75 AND all totals correct $(69, 123, 108, 300)$ **[B1]**

(b) Girls who do not choose Drama $= 45 + 75 = 120$ **[M1]**

$\frac{120}{300}$ or $\frac{2}{5}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

16 **(a)** Correct Managers row: 48, 12, 60 **[B1]**

Correct Office Workers row: 90, 60, 150 **[B1]**

Correct Technicians row: 78, 12, 90 **[B1]**

Correct Total row: 216, 84, 300 **[B1]**

(b) Denominator of 84 seen **[M1]**

$\frac{60}{84}$ or $\frac{5}{7}$ **[A1]**

(c) $\frac{12}{60} \times \frac{48}{59}$ **[M1]**

$\frac{12}{60} \times \frac{48}{59} + \frac{48}{60} \times \frac{12}{59}$ or $2 \times \left(\frac{12}{60} \times \frac{48}{59}\right)$ **[M1]**

$\frac{96}{295}$ or 0.325 (0.3254...) **[A1]**

[Back to Question] *[Total Marks: 9]*

17 **(a)** $\frac{3}{4}x$ or $0.75x$ **[B1]**

(b) Forms correct equation: $\frac{3}{4}x + \left(\frac{1}{4}x + 30\right) = 110$ **[M1]**

Simplifies to $x + 30 = 110$ leading to $x = 80$ with no errors **[A1]**

(c) Male row correct: 20 (Golf), 60 (Tennis), 80 (Total) **[B1]**

Female row correct: 70 (Golf), 50 (Tennis), 120 (Total) **[B1]**

Total row correct: 90 (Golf), 110 (Tennis), 200 (Total) **[B1]**

(d) Identifies correct fraction for first female: $\frac{70}{90}$ **[M1]**

Multiplies by correct fraction for second female: $\frac{70}{90} \times \frac{69}{89}$ **[M1]**

$\frac{4830}{8010}$ or $\frac{161}{267}$ **[A1]**

[Back to Question] *[Total Marks: 9]*

18 **(a)** Total number of left-handed students $= 3x$ **[B1]**

$\frac{3x}{120} \times \frac{3x-1}{119} = \frac{3}{34}$ **[M1]**

$\frac{9x^2 - 3x}{14280} = \frac{3}{34}$ or $3x(3x-1) = 1260$ **[M1]**

$9x^2 - 3x = 1260 \implies 3x^2 - x - 420 = 0$ (with no errors seen) **[A1]**

(b) $(3x - 36)(x + 11.66...) = 0$ or correct substitution into quadratic formula **[M1]**

$x = 12$ **[A1]**

Total students equation: $x + y + 2x + y - 6 = 120$ **[M1]**

$3(12) + 2y - 6 = 120 \implies 2y = 90 \implies y = 45$ **[A1]**

(c) Total right-handed $= 2y - 6 = 2(45) - 6 = 84$ OR Right-handed girls $= 45 - 6 = 39$ **[M1]**

$\frac{39}{84}$ or $\frac{13}{28}$ **[A1]**

[Back to Question] *[Total Marks: 10]*